



BriteTest

Runner Reference

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This document provides a reference to the Runner API. It includes types, structs, unions, enums, macros, and functions. Additionally, it provides a reference to Runner Framework concepts.

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Preface

This document is intended for BriteTest user and contributors who need a reference for the BriteTest Runner framework and API.

For a list of other BriteTest documents and the repository layout, see the Documentation Guide (`Documentation_Guide.md`).

For a glossary of terms, see the Glossary Reference (`Glossary_Reference.md`).

Document Version History

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- The **Document** column records the document's version with the format **M.u** (Major, update).
- The **Runner** column records the Runner API version current at the time this document version was published and is defined by its `RA_RUNNER_VERSION` macro.
- The **Test** column records the Test API version current at the time this document version was published and is defined by its `RA_TEST_VERSION` macro.
- Both Runner and Test use the version format "**M.m.p**" (Major, minor, patch).
- **M** is the same for the Document, Runner, and Test versions.

The document's update version tracks released updates to this document and does not correspond to a minor or patch version. **u** increments when document changes are published in a release without a change to **M**, and it resets to 0 when **M** is incremented.

Table of Contents

1. Introduction	1
2. Types	2
2.1. ra_resubt_t	??
2.2. ra_state_t	??
3. Enums	3
3.1. ra_exit_code_t	3
3.2. ra_return_code_t	3
4. Macros for Limits	5
5. Macros for the Orchestrator Function	6
5.1. RA_DECLARE_ORCHESTRATOR	6
5.2. RA_INIT_ORCHESTRATOR	6
5.3. RA_PARSE_ARGS	6
5.4. RA_OPEN_REPORT	7
5.5. RA_WRITE_RESULT	7
5.6. RA_CLOSE_REPORT	7
5.7. RA_EXIT	7
6. Macros for a Test Group Function	8
6.1. RA_DECLARE_GROUP	8
6.2. RA_INIT_GROUP	8
6.3. RA_RETURN	8
7. Macros to Execute a Test Group or Test	??
7.1. RA_GROUP	9
7.2. RA_TEST	9
8. Macros for Concurrent Blocks	10
8.1. RA_BEGIN_CONCURRENT	10
8.2. RA_END_CONCURRENT	10
9. Macros for Versioning	11
9.1. RA_RUNNER_VERSION	11
9.2. RA_VERSION_MAJOR	11
9.3. RA_VERSION_MINOR	11
9.4. RA_VERSION_PATCH	11

9.5. RA_VERSION_NUM	11
9.6. RA_VERSION_HEX	12
9.7. RA_VERSION_CMP	12
10. Macros for Customization	14
10.1. RA_PRINT_ERR_HELP	13
11. Functions for Customization	14
11.1. ra_funcname	14
11.2. ra_print_note	14
12. Test Command Line	15
13. Output and Golden Files	16
12.1. Output Files	17
12.2. Golden Files	18
12.3. Output/Golden Rationale	??

1. Introduction

TODO: add introduction, common rules.

2. Types

2.1. ra_resubt_t

Type for result counters.

Note: A test group function or test with a fault sets its result with the fault count set to `SIZE_MAX` to indicate a function-level fault.

See also: - `ra_currentresult(void)` in Customization Helper Functions. - `RA_GROUP(...)` notes on fault capture semantics and how function-level faults are represented.

```
typedef struct
{
    size_t pass;
    size_t fail;
    size_t fault;
    size_t injected_fail;
    size_t injected_fault;
} ra_resubt_t;
```

2.2 ra_state_t

Type for maintaining the state of the orchestrator (main) or a test function.

```
typedef struct
{
    size_t id;
    char *funcname;
    size_t current_level;
    size_t groupid;
    char *groupname;
    int isolation; // 0 none, 1 thread, 2 process.
    size_t category_id;
    size_t num_results_merged;
    size_t pass;
    size_t fail;
    size_t fault;
    size_t injected_fail;
    size_t injected_fault;
    size_t total_pass;
    size_t total_fail;
    size_t total_fault;
    size_t total_injected_fail;
    size_t total_injected_fault;
    ra_state_t parent;
    ra_state_t prev;
    ra_state_t next;
} ra_state_t;
```

3. Enums

3.1. ra_exit_code_t

Exit codes are grouped into classes: - Normal exit (*if* not a test run request): RA_EXIT[_*] (0-99). - Test outcome (*if* a test run request): RA_TEST[_]* (0-99). - Fatal usage (of framework): RA_FATAL_USAGE[_*] (100-199). - Fatal internal (in framework): RA_FATAL_INTERNAL[_*] (200-299). - Fatal system (call failed): RA_FATAL_SYSTEM[_*] (300-399).

Fatal codes indicate that the runner could not complete normal or test execution.

RA_FATAL_USAGE, RA_FATAL_INTERNAL, and RA_FATAL_SYSTEM are general codes for when a more specific code does not apply for the class.

typedef enum

```
{
    // Normal Exit (*if* not a test run request): 0-99.
    RA_EXIT_OK = 0,

    // Test Outcome (*if* a test run request): 0-99.
    RA_TEST_PASS = 0,
    RA_TEST_FAIL = 1,
    RA_TEST_FAULT = 2,
    RA_TEST_FAIL_FAULT = 3,

    // Fatal Usage (of framework): 100-199.
    RA_FATAL_USAGE = 100,
    RA_FATAL_USAGE_INVALID_ARG = 101,
    RA_FATAL_USAGE_INVALID_STATE = 102,

    // Fatal Internal (in framework): 200-299,
    RA_FATAL_INTERNAL = 200,
    RA_FATAL_INTERNAL_ASSERT = 201,
    RA_FATAL_INTERNAL_INVARIANT = 202,

    // Fatal System (call failed) 300-399.
    RA_FATAL_SYSTEM = 300,
    RA_FATAL_SYSTEM_OPEN = 301,
    RA_FATAL_SYSTEM_READ = 302,
    RA_FATAL_SYSTEM_WRITE = 303,
    RA_FATAL_SYSTEM_FORK = 304,
    RA_FATAL_SYSTEM_THREAD = 305
} ra_exit_code_t;
```

3.1. ra_return_code_t

Return codes are for functions that return an integer (e.g., `int`) and are grouped into classes: - Success - Compare - Boolean - RA_INVALID[_*] invalid usage codes (invalid arguments). - RA_SYSTEM[_*] failed system codes (OS/runtime/resource failures).

RA_INVALID and RA_SYSTEM) are general codes for when a more specific code does not apply for the class.

```
typedef enum
{
    // Success
    RA_OK = 0,

    // Compare
    RA_LESS = -1,
    RA_EQUAL = 0,
    RA_GREATER = 1,

    // Boolean
    RA_FALSE = 0,
    RA_TRUE = 1,

    // -100 to -199: Invalid Usage.
    RA_INVALID = -100,
    RA_INVALID_ARG = -101,
    RA_INVALID_ARG_VERSION = -102,
    RA_INVALID_ARG_TOO_LONG = -103,

    // -300 to -399: Failed system call.
    RA_SYSTEM = -300,
    RA_SYSTEM_OPEN = -301,
    RA_SYSTEM_READ = -302,
    RA_SYSTEM_WRITE = -303,
    RA_SYSTEM_FORK = -304,
    RA_SYSTEM_THREAD = -305
} ra_return_code_t;
```


4. Macros for Limits

Maximum values for path length, filename length, and guard levels.

```
#define RA_MAX_PATH_LEN ((size_t)4096) #define RA_MAX_FILENAME_LEN  
((size_t)255) #define RA_MAX_LEVEL ((size_t)32)
```

Note: The limit of 32 levels is unlikely to be exceeded if there are 2 or more `RA_GROUP` or `RA_TEST` macros at each level. It is expected that a level will generally have 2 or more per level.

5. Macros for the Orchestrator Function

Macros used to define or declare the Orchestrator (`main`) function

5.1. `RA_DECLARE_ORCHESTRATOR(funcname) [;]`

A semicolon is not allowed if the macro is followed by its definition `{...}` of the orchestrator function; otherwise, it is required and indicates this is a declaration and a forward-reference to the orchestrator function.

funcname: - Token specifying `main`.

Fatal exit: - `FATAL_USAGE_DECLARE_ORCHESTRATOR` - `FATAL_INTERNAL` - `FATAL_SYSTEM`

5.2. `RA_INIT_ORCHESTRATOR(funcname, id, project, size_t maxparallel);`

funcname: - Token specifying `main`.

id: - A pointer to a string specifying only alphanumeric characters which is used to identify the orchestrator function. - Default is `"o"` *if* `id` is `NULL` pointer or its string is empty. - A short `id` (not exceeding 3 characters) is recommended. - The `id` must be unique for the orchestrator and test group functions; a violation causes the test runner to terminate.

project: - A token identifying the project. - The token is used to generate a default report title if one is not set by the `RA_PARSE_ARGS(...)` macro.

maxparallel: Maximum number of `RA_GROUP` and `RA_TEST` macros that are allowed to execute in parallel.

Fatal exit: - `FATAL_USAGE_INIT_ORCHESTRATOR` - `FATAL_INTERNAL` - `FATAL_SYSTEM`

5.3. `RA_PARSE_ARGS(size_t maxargs, char *customflags, char *defaultreportfilename)`

maxargs: - The maximum number arguments in the command line. - The value must 2 or greater. - The first 2 arguments and BriteTest flags are parsed by the macro. - Additional arguments must be parsed by customization code.

customflags: - A string of custom flags (e.g., `"-l-file-Q"`). - The macro causes the test runner to terminate if a non-BriteTest flag is not in the string. - The actual parsing of these must be customization code.

defaultreportfilename: - A string defining a default report file name without an extension. The extension is `.md` since the report uses Markdown Document formatting. - The maximum length including a null terminator character is `MAX_FILENAME_LEN` (129),

Termination Exit Codes: - `RA_ARG_ERROR` - `RA_FLAG_ERROR`

See “Test Command Line” for details on command-line argument and flags.

5.4. RA_OPEN_REPORT(char *title);

title: - A string defining the report title. - If **title** is NULL or empty, a generated default report title “ Test Report” is used. - The maximum length including a null terminator character is **MAX_TITLE_LEN** (129),

Termination Exit Codes: - **RA_OPEN_ERROR**

5.5. RA_WRITE_RESULT(gtm, char *category);

gtm: - A **RA_GROUP(...)** or **RA_TEST(,,)** macro.

category: - A pointer to a string with the name or description of the test category. - The maximum length including a null terminator character is **MAX_CATEOGRY_LEN** (129).

Termination Exit Codes: **RA_WRITE_ERROR**.

5.6. RA_CLOSE_REPORT(char *notes);

notes: - A string of note lines to append to the test report. - Each note line must end with '\n' (newline). - If **notes** is NULL or empty, there are no notes to append. - The maximum length including a null terminator character is **MAX_NOTE_LEN** (10001).

Termination Exit Codes: - **RA_CLOSE_ERROR**

The macro writes the totals, pre-defined explanatory notes, **notes**, notes from the temporary notes file, and final summary line to the test report file, and then closes the test report.

5.7. RA_EXIT;

Exit codes: - **RA_PASS** (0) - **RA_FAIL** (1) - **RA_FAULT** (2) - **RA_FAIL_FAULT** (3)

Termination Exit Codes: - **RA_EXIT_ERROR**

6. Macros for a Test Group Function

Macros used to define or declare a test group function.

6.1. **RA_DECLARE_GROUP(funcname)**

funcname: - Token specifying a function name other than **main**.

A semicolon is not allowed if the macro is followed by its definition `{...}` of the test group function; otherwise, it is required and indicates this is a declaration and a forward-reference to the test group function.

Termination Exit Codes: - **RA_DECLARE_GROUP_ERROR**

6.2. **RA_INIT_GROUP(funcname, id, maxparallel)**

funcname: - The same token as the specified token for **funcname** in the preceding **RA_DECLARE_GROUP** macro defining the containing test group function.

id: - A pointer to a string specifying only alphanumeric characters which is used to identify the test group function. - Default is "**funcname**" if **id** is NULL pointer or its string is empty.
- A short id (not exceeding 3 characters) is recommended. - **id** must be unique for the orchestrator and test group functions; a violation causes the test runner to terminate.

maxparallel: - Maximum number of **RA_GROUP** and **RA_TEST** macros that are allowed to execute in parallel.

Termination Exit Codes: - **RA_INIT_GROUP_ERROR**

6.3. **RA_RETURN**

Termination Exit Codes: - **RA_RETURN_ERROR**

7. Macros to Execute a Test Group or a Test

- `RA_GROUP(funcname, id, [include], [isolation])[;]`
- `RA_TEST(expression, id, [include], [isolation])[;]`

A semicolon is not allowed if the macro is used as an argument to the `RA_WRITE_RESULT` macro; otherwise, it is required.

id: - A pointer to a string specifying only alphanumeric characters for the test group or test.
 - The value must be unique within the containing orchestrator or test group function - A short id (not exceeding 3 characters) is recommended or use the default. - If `id` is a NULL pointer or its string is empty, default is `n`, where `n` is 1 for the first `RA_GROUP` or `RA_TEST` macro where `id` is NULL or empty, and incremented by 1 for each subsequent `RA_GROUP` and `RA_TEST` macro where `id` is a NULL pointer or its string is empty. - See the `RA_ID` macro for obtaining the `id` for the next generated id.

include: - A single character token or omit (defaults to 1): - 0 — Never execute. This is useful to disable a test (e.g., failing or faulting test for which the fix has been deferred). - 1 — Always execute (e.g., smoke tests). - 2-9 — Execute only when the `-In` flag is specified in the test command line and `n >= include`. - I — Execute only when the `-I` flag is specified in the test command line. This is useful for injecting a fail/fault for verifying behavior and report output if fails/faults were to occur. See also `RA_FAIL` and `RA_FAULT(type)` for simulating a fail or fault, respectively.

See “Test Command Line” for details on command line flags.

isolation: - Single character token or omit (defaults to 0): - 0 — Same thread. - 1 — Separate thread. - 2 — Separate process.

See “Isolation” in the Runner Guide.

7.1. `RA_GROUP(funcname, id, [include], [isolation])[;]`

Termination Exit Codes: - `RA_GROUP_ERROR`

7.2. `RA_TEST(expression, id, [include], [isolation])[;]`

Termination Exit Codes: - `RA_TEST_ERROR`

8. Macros for Concurrent Blocks

8.1. `RA_BEGIN_CONCURRENT(blockname)`

blockname: - Token specifying the name of the block. - Token must be the same as for the subsequent matching `RA_END_CONCURRENT(blockname)`.

Termination Exit Codes: - `RA_BEGIN_GROUP_ERROR`

8.2. `RA_END_CONCURRENT(blockname);`

blockname: - Token specifying the name of the block. - Token must be the same as for the preceding matching `RA_BEGIN_CONCURRENT(blockname)`.

Termination Exit Codes: - `RA_END_GROUP_ERROR`

9. Macros for Versioning

9.1. RA_RUNNER_VERSION

Value: - BriteTest Runner's version as a literal string of type `char[n]`, where `n` is between 5 and 9 (including the terminating null character). - Format: `M.m.p`, where `M`, `m`, and `p` are one or two digits. - `M` is the major version, `m` is the minor version, and `p` is the patch version.

The major version is incremented for major additions, removal of deprecated features, or unavoidable incompatible API changes.

The minor version is incremented for backward-compatible additions or deprecating features.

The patch version is incremented for bug fixes or internal improvements.

Examples: - `RA_RUNNER_VERSION` $\rightarrow$ "1.0.9" - `RA_RUNNER_VERSION` $\rightarrow$ "1.3.11" - `RA_RUNNER_VERSION` $\rightarrow$ "12.05.18"

9.2. RA_VERSION_MAJOR(v)

`v`: pointer to a version string.

Value: - The major version in `v` cast to `int`. - Value is `RA_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Examples: - `RA_VERSION_MAJOR("5.0.9")` $\rightarrow$ (int)5 - `RA_VERSION_MAJOR("5.000.9")` $\rightarrow$ `RA_INVALID_VERSION`

9.3. RA_VERSION_MINOR(v)

`v`: pointer to a version string.

Value: - The minor version in `v` cast to `int`. - Value is `RA_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Examples: - `RA_VERSION_MINOR("5.00.9")` $\rightarrow$ (int)0 - `RA_VERSION_MINOR("5.1.001")` $\rightarrow$ `RA_INVALID_VERSION`

9.4. RA_VERSION_PATCH(v)

`v`: pointer to a version string.

Value: - The patch version in `v` cast to `int`. - Value is `RA_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Examples: - `RA_VERSION_PATCH("5.0.12")` $\rightarrow$ (int)12 - `RA_VERSION_PATCH("5.000.9")` $\rightarrow$ `RA_INVALID_VERSION`

9.5. RA_VERSION_NUM(v)

`v`: pointer to a version string.

Value: - Representation of the version `v` cast to `int`. - Value is `RA_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Form: MMmmpp for comparisons, e.g., 10000 for version 1.0.0, 10200 for version 1.2.0, or 11212 for version 1.12.12.

Examples: - `RA_VERSION_NUM("1.0.9")` $\rightarrow$ `(int)0x010009` - `RA_VERSION_NUM("2.5.3")` $\rightarrow$ `(int)0x020503`

9.6. `RA_VERSION_HEX(v)`

v: - Pointer to a version string.

Value: - Hex representation of version `v` cast to `int`. - Value is `RA_INVALID_VERSION` `((int)-2)` *if* `v` is an invalid version string.

Hexadecimal form 0xMMmmpp for display/debugging, e.g., 0x010000 for version 1.0.0, 0x010200 for version 1.2.0, or 0x011212 for version 1.12.12.

Examples: - `RA_VERSION_HEX("1.0.9")` $\rightarrow$ `0x010009` - `RA_VERSION_HEX("2.5.3")` $\rightarrow$ `0x020503`

9.7. `RA_VERSION_CMP(v1, v2)`

v1: - Pointer to a version string.

v2: - Pointer to a version string.

Value: - Integer (`int`) result comparing version `v1` to `v2`: - `RA_EQUAL` `((int)0)` — equal - `RA_LESS` `((int)-1)` — `v1` is less than `v2` - `RA_GREATER` `((int)1)` — `v1` is greater than `v2` - `RA_INVALID_VERSION` `((int)-2)` — invalid `v1` or invalid `v2` - Leading zeros in `M`, `m`, and `p` are ignored during comparison.

Examples: - `RA_VERSION_CMP(RA_RUNNER_VERSION, "1.10.0")` $\rightarrow$ `RA_GREATER` (*if* `RA_RUNNER_VERSION` is `"1.10.6"`) - `RA_VERSION_CMP(RA_RUNNER_VERSION, "1.10.0")` $\rightarrow$ `RA_LESS` (*if* `RA_RUNNER_VERSION` is `"1.8.0"`) - `RA_VERSION_CMP(RA_RUNNER_VERSION, "1.10.0")` $\rightarrow$ `RA_EQUAL` (*if* `RA_RUNNER_VERSION` is `"1.10.0"`) - `RA_VERSION_CMP("1.10.0x", "1.10.0")` $\rightarrow$ `RA_INVALID_VERSION` - `RA_VERSION_CMP("1.10.0", "1.10y.0")` $\rightarrow$ `RA_INVALID_VERSION`

10. Macros for Customization

10.1. `RA_PRINT_ERR_HELP(char *err, char help)`

Print err and help text to stdout. - `err` is prefixed with `ra_err_prefix()`. - Help text is formed using `ra_usage()` and `ra_help()`.

err - Pointer to error string.

help - 0: don't print help text. - non-zero: print help text.

Fatal exit: - `FATAL_USAGE` - `FATAL_INTERNAL` - `FATAL_SYSTEM`

Note: Use `ra_set_err_prefix()` to set the prefix.

Note: Use `ra_set_usage()` and/or `ra_set_help()` to define the help text.

11. Functions for Customization

11.1. int ra_funcname(char *funcname, size_t outlen)

11.2. int ra_print_note(const char *notes)

Opens a temporary file if not already open and writes the notes to the temporary file.

notes: - A string of note lines to write to the temporary file. - Each note line must end with '\n' (newline). - If **notes** is NULL or empty, the temporary file is removed if it exists. - The maximum length including a null terminator character is MAX_NOTE_LEN (10001).

Return: - RA_SUCCESS - RA_NOTES_REMOVED - RA_NOTES_OPEN_ERROR - RA_PRINT_NOTE_ERROR

void ra_current_time(char *current_time, size_t size)

Get the current time as a formatted string.

current_time: - Buffer to store the formatted time string.

size Size of the buffer.

Note: On error, current_time is set to “unknown time”.

Customization: Get and set.

```
char *ra_executable_name(void) void ra_set_executable_name(char *en) char
*ra_defautb_dirpath(void) void ra_set_defautb_dirpath(char *dp) char *ra_err_prefix(void)
void ra_set_err_prefix(char *pe) char *ra_args_options(void) void ra_set_args_options(char
*ao) char *ra_usage(void) void ra_set_usage(char *u) char *ra_help(void) void
ra_set_help(char *h)
```

Customization Helper Functions

```
size_t ra_currentlevel(void) ra_resubt_t ra_currentresult(void) ra_total_t
ra_currenttotal(void) size_t ra_maxparallel(size_t level) size_t ra_currentparallel(void)
int ra_isisolated(void) int ra_isthreadisolated(void) int ra_isprocessisolated(void)
size_t ra_groupid(void) char *ra_groupname(void)
```

```
char *ra_project(void) size_t ra_maxargs(void) char *ra_title(void) size_t
ra_categoryid(void) char *ra_category(void) char *ra_funcname(void) char
*ra_notes(void) char *ra_assertexpression(void) char ra_inject(void) char
ra_isolation(void) char ra_orchestrator(void) char ra_testfunction(void) char
ra_assert(void)
```

```
char *ra_dirpath(void) char *ra_filepath(void) char *ra_filename(void)
```

Return: 1 true, 0 false int ra_isfilename(char *name)

Return: 1 true, 0 false, -101 invalid directory path. int ra_isreaddirpath(char *path)
int ra_iswritedirpath(char *path)

Return 1 true, 0 false, -103 invalid file path. int ra_isreadfilepath(char *path) int
ra_iswritefilepath(char *path)

12. Test Command Line

TODO.

13. Output and Golden Files

Output files in `tests/output/` and golden files in `tests/golden/` subdirectories of the repository root provide a directory layout for capturing and validating test output.

13.1. Output Files

A copy of a named output file is captured to `tests/output/` : - If the output file name has the form `.`, the `tests/output/` file is named `-.ext`. - If the output file name has the form (with no extension), the `tests/output/` file is named `-`.

A copy of `stdout` is captured in `tests/output/` output file named `stdout-<uid>` and a copy of `stderr` is captured as an output file named `stderr-<uid>`.

`<uid>` is a unique id string for each captured output file within a test run. It is used to avoid filename collisions when the same output file name appears multiple times (e.g., tests of a command line varying options each generationg `stdout` and `stderrj`).

Temporary files are not captured in `tests/output/`.

A metadata file (in Markdown Document format) for each output file is also written to `tests/output/`: - If the output file name has the form `.`, the `tests/output/` metadata file is named `-.ext.md`. - If the output file name has the form (with no extension), the `tests/output/` file is named `-.md`.

An output file and its metadata file in `tests/output/` may be copied (promoted) to the `tests/golden/` subdirectory of the repository root directory. See the following section.

13.2. Golden Files

A metadata golden file having a name with the form `.ext` applies to all golden files named `-ext` and can be added manually or by customization code added to the test runner.

A metadata golden file having a name with the form `.md` applies to all golden files named `-` with no extension and can be added manually or by customization code added to the test runner.

Golden files (located in the `tests/golden/`) are handled as follows:

- If the corresponding golden file with the same name does not exist,
 - If a metadata golden file `.ext` exists and contains the line `## MATCH`, comparison is skipped and a match is forced. If it contains the line `## MISMATCH`, comparison is skipped and a mismatch is forced. Additional markdown text can follow either marker to document why the outcome is forced for that file or set of files.
 - Otherwise, the output file is automatically copied (promoted) into the `tests/golden` directory and a match forced.
 - The test report notes that the file was promoted.
- If the corresponding golden file exists and contains the line `## MATCH`, comparison is skipped and a match is forced.
- If the corresponding golden file exists and contains the line `## MISMATCH`, comparison is skipped and a mismatch is forced.
- Otherwise, the output file is compared with its existing golden file of the same name in the `tests/golden` directory to determine whether they match.

If an output file does not match its golden file but inspection determines the output is correct, update the golden file with a copy of the output. Alternatively, remove the golden file so that the next run will automatically promote the output.

13.3. Output/Golden Rationale

BriteTest uses a parallel `tests/golden` directory with golden files that keep the exact same filename as their corresponding test output. This design avoids the common problems found in extension-based naming schemes:

- **No filename collisions** Different outputs such as `foo.c` and `foo.h` remain distinct (`foo-<uid>.c` vs. `foo-<uid>.h`) instead of collapsing into a single `foo.golden`.
- **No special-case rules** The golden filename is always identical to the output filename.
- **Preserves file type information** Because the original extension is retained, it is immediately clear whether a golden file contains text, Markdown Document, JSON, HTML, binary data, or something else.
- **Simpler mental model** Developers do not need to remember naming conventions or extension-replacement rules. Golden files simply live in a parallel directory and share the same name as the output they validate.